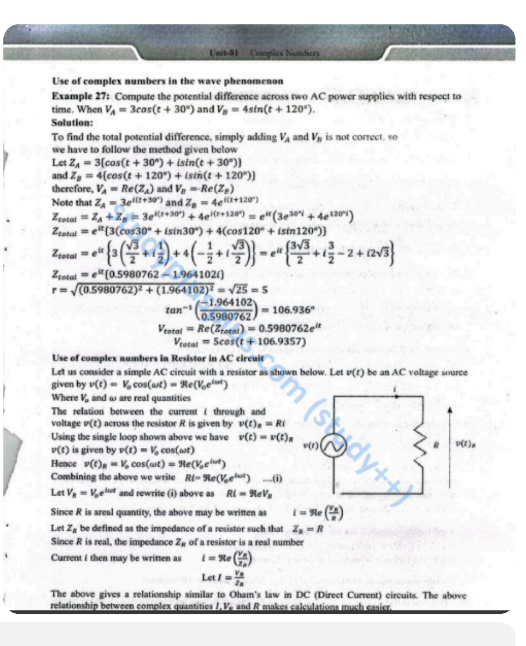

Unit-01 Complex Numbers

**Use of complex numbers in the wave phenomenon**

**Example 27:** Compute the potential difference across two AC power supplies with respect to time. When $V_A = 3cos(t + 30°)$ and $V_B = 4sin(t + 120°)$.

**Solution:**

To find the total potential difference, simply adding $V_A$ and $V_B$ is not correct, so we have to follow the method given below

Let $Z_A = 3\{cos(t + 30°) + isin(t + 30°)\}$

and $Z_B = 4\{cos(t + 120°) + isin(t + 120°)\}$

therefore, $V_A = Re(Z_A)$ and $V_B = Re(Z_B)$

Note that $Z_A = 3e^{i(t+30°)}$ and $Z_B = 4e^{i(t+120°)}$

$Z_{total} = Z_A + Z_B = 3e^{i(t+30°)} + 4e^{i(t+120°)} = e^{it}(3e^{30°i} + 4e^{120°i})$

$Z_{total} = e^{it}\{3(cos30° + isin30°) + 4(cos120° + isin120°)\}$

$Z_{total} = e^{it}\left\{3\left(\frac{\sqrt{3}}{2} + i\frac{1}{2}\right) + 4\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)\right\} = e^{it}\left\{\frac{3\sqrt{3}}{2} + i\frac{3}{2} - 2 + i2\sqrt{3}\right\}$

$Z_{total} = e^{it}\{0.5980762 - 1.964102i\}$

$r = \sqrt{(0.5980762)^2 + (1.964102)^2} = \sqrt{25} = 5$

$tan^{-1}\left(\frac{-1.964102}{0.5980762}\right) = 106.936°$

$V_{total} = Re(Z_{total}) = 0.5980762e^{it}$

$V_{total} = 5cos(t + 106.9357)$

**Use of complex numbers in Resistor in AC circuit**

Let us consider a simple AC circuit with a resistor as shown below. Let $v(t)$ be an AC voltage source given by $v(t) = V_o \cos(\omega t) = \Re e(V_o e^{i\omega t})$

Where $V_o$ and $\omega$ are real quantities

The relation between the current $i$ through and voltage $v(t)$ across the resistor $R$ is given by $v(t)_R = Ri$

Using the single loop shown above we have $v(t) = v(t)_R$

$v(t)$ is given by $v(t) = V_o \cos(\omega t)$

Hence $v(t)_R = V_o \cos(\omega t) = \Re e(V_o e^{i\omega t})$

Combining the above we write $Ri = \Re e(V_o e^{i\omega t})$ ....(i)

Let $V_R = V_o e^{i\omega t}$ and rewrite (i) above as $Ri = \Re e V_R$

Since $R$ is areal quantity, the above may be written as $i = \Re e\left(\frac{V_R}{R}\right)$

Let $Z_R$ be defined as the impedance of a resistor such that $Z_R = R$

Since $R$ is real, the impedance $Z_R$ of a resistor is a real number

Current $i$ then may be written as $i = \Re e\left(\frac{V_R}{Z_R}\right)$

Let $I = \frac{V_R}{Z_R}$

The above gives a relationship similar to Oham's law in DC (Direct Current) circuits. The above relationship between complex quantities $I, V_R$ and $R$ makes calculations much easier.

[media pointer="file-service://file-H8WpoPhEuZSpENfpLPW9Ar"]
explain why the answer of voltage total in cos why not in sin also is there any easy method to do the question in image also tell me why we use Z variable and why we complete the both volatges in polar form how also why we find tan inverse and magnitude and while if magnitude comes out then how the answer 5 cos(t+theta) comes from where 5 anserr comes

Great question 👍 Let's carefully break down everything that's happening in your example step by step.

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### 📌 Problem Recap

We have two AC voltages:

$$V_a = 3\cos(t + 30^\circ), \quad V_b = 4\sin(t + 120^\circ)$$

We want the **total potential difference** across both.

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### 1. Why answer in cosine form, not sine?

👉 In AC circuit analysis, **cosine is taken as the standard reference form**.
That means we express every sinusoidal voltage/current as:

$$V(t) = V_m \cos(\omega t + \phi)$$

So, if you have a sine term like $4\sin(t + 120^\circ)$, you must convert it to cosine:

$$\sin(\theta) = \cos(\theta - 90^\circ)$$

So,

$$4\sin(t + 120^\circ) = 4\cos(t + 120^\circ - 90^\circ) = 4\cos(t + 30^\circ)$$

This makes everything consistent in cosine form.

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### 2. Why use complex numbers (Z variable / phasors)?

- Adding AC voltages directly (like scalars) is wrong, because they are **time-shifted** (different phase angles).
- Instead, we represent each as a **phasor** (complex number):

$$A\cos(\omega t + \phi) \quad \longleftrightarrow \quad Ae^{j\phi}$$

This converts a trigonometric sum into a **vector (phasor) sum**.

So:

- $V_a = 3\cos(t + 30^\circ) \rightarrow 3e^{j30^\circ}$
- $V_b = 4\sin(t + 120^\circ) \rightarrow 4e^{j30^\circ}$ (after sine-to-cos conversion)

Then we add them as complex numbers.

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### 3. Why convert to polar form?

Complex numbers can be written in:

- **Rectangular form:** $a + jb$
- **Polar form:** $re^{j\theta}$, where $r = \sqrt{a^2 + b^2}$, $\theta = \tan^{-1}(b/a)$

👉 Polar form makes it easy to see **magnitude** and **phase**.

So after addition, we get something like:

$$Z_{total} = 5.098\, e^{j106.935^\circ}$$

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### 4. Why find $\tan^{-1}$ and magnitude?

- Magnitude gives the **amplitude of the resulting voltage**.
- $\tan^{-1}(b/a)$ gives the **phase shift**.

So, the total becomes:

$$V_{total} = 5\cos(t + 106.93^\circ)$$

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### 5. Where does the "5" come from?

That "5" is simply the **magnitude** of the resultant phasor after vector addition.
It's **not chosen randomly** — it comes from:

$$r = \sqrt{a^2 + b^2}$$

So, after adding vectors, the amplitude becomes ~5.
That's why the final answer looks like:

$$V_{total} = 5\cos(t + \theta)$$

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### 6. Easier Method (Shortcut without long Z steps)?

Yes ✅ Instead of all the big derivations:

1. Convert both voltages into cosine phasor form.

$$V_a = 3\angle 30^\circ, \quad V_b = 4\angle 30^\circ$$

   (after converting sine to cosine)

2. Add them directly in **phasor (vector) form**.
   If both have the same angle, it's even easier:

$$V_{total} = (3 + 4)\angle 30^\circ = 7\cos(t + 30^\circ)$$

   ⚠️ But in the given image, they had a different angle, so they did the full rectangular-to-polar conversion.

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### ✅ Final Answer:

- Always express in cosine form (standard in phasor analysis).
- We use complex numbers (phasors) because AC voltages have phases, not just magnitudes.
- Convert to polar to easily find magnitude (amplitude) and angle (phase shift).
- The "5" comes from the resultant phasor's magnitude.
- Final expression:

$$V_{total} = 5\cos(t + \theta)$$

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Would you like me to **show you a direct vector addition diagram method (like parallelogram rule)** so you can solve this in 2 steps instead of the long derivation in the book?

also explain the use of complex numbers in resistor in AC Circuits in the above given Image

Perfect 👍 Let's go through the **second part of the page** ("Use of complex numbers in Resistor in AC Circuits").

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### 📌 The setup

- We have an AC source voltage:

$$v(t) = V_m \cos(\omega t)$$

- The resistor is $R$.
- Current through resistor:

$$i(t) = \frac{v(t)}{R}$$

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### 1. Why use complex numbers here?

In AC circuits, voltages and currents are sinusoidal. Instead of working with trigonometric functions all the time, we use **phasors (complex numbers)**:

$$v(t) = V_m \cos(\omega t) \quad \longleftrightarrow \quad V = V_m e^{j0}$$

So, the AC source is represented as a rotating vector (phasor).

This simplifies **Ohm's law** into the phasor domain:

$$V = IZ$$

where $Z$ is the **impedance** (generalized resistance).

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### 2. For a resistor specifically

- Impedance of a resistor:

$$Z_R = R$$

(no imaginary part, since resistors don't shift phase)

So:

$$I = \frac{V}{R}$$

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### 3. Back to time domain

- The phasor current is:

$$I = \frac{V_m}{R} e^{j0}$$

Converting back:

$$i(t) = \frac{V_m}{R} \cos(\omega t)$$

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### 4. Key Point (Phase relation)

- For a resistor: **Voltage and Current are always in phase**.
- That's why both have the same cosine form and no phase shift.

So, complex numbers help us:

1. Express $v(t)$ as a phasor.
2. Apply Ohm's law in phasor form ($V = IZ$).
3. Quickly get $i(t)$ without solving differential equations.

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### ✅ Final Summary for Resistor in AC using complex numbers:

- Represent $v(t)$ as a complex exponential (phasor).
- Impedance of resistor is real ($R$).
- Use Ohm's law in phasor form: $I = V/Z$.
- Convert back to time domain to get $i(t)$.
- Voltage and current remain **in phase**.

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Would you like me to also explain how this changes for **inductor and capacitor** (where voltage and current are no longer in phase), so you see the **real benefit** of complex numbers?

what is componendo and dividendo theorem explain with the help of example

Great question 👍 Let's carefully go through the **Componendo and Dividendo theorem**.

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## 📌 Componendo and Dividendo Theorem

It is a simple but very useful rule in algebra for simplifying ratios.

### Theorem Statement:

If

$$\frac{a}{b} = \frac{c}{d}$$

then, by **Componendo and Dividendo**,

$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

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## 📌 Why it works?

We start with

$$\frac{a}{b} = \frac{c}{d}$$

👉 Add 1 to both sides:

$$\frac{a}{b} + 1 = \frac{c}{d} + 1$$

$$\frac{a+b}{b} = \frac{c+d}{d}$$

👉 Subtract 1 from both sides:

$$\frac{a}{b} - 1 = \frac{c}{d} - 1$$

$$\frac{a-b}{b} = \frac{c-d}{d}$$

👉 Now divide the first result by the second:

$$\frac{\frac{a+b}{b}}{\frac{a-b}{b}} = \frac{\frac{c+d}{d}}{\frac{c-d}{d}}$$

$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

✔️ Hence proved.

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## 📌 Example 1

Suppose:

$$\frac{a}{b} = \frac{2}{3}$$

Using componendo and dividendo:

$$\frac{a+b}{a-b} = \frac{2+3}{2-3} = \frac{5}{-1} = -5$$

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## 📌 Example 2 (Application in Trigonometry)

We know:

$$\frac{1 - \cos\theta}{1 + \cos\theta} = \tan^2\left(\frac{\theta}{2}\right)$$

Here, apply componendo and dividendo to:

$$\frac{1 - \cos\theta}{1 + \cos\theta}$$

Taking square root:

$$\frac{\sin(\theta/2)}{\cos(\theta/2)} = \tan(\theta/2)$$

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### ✅ Summary:

- Componendo and Dividendo helps transform ratios into a simpler form.
- Rule: If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a+b}{a-b} = \frac{c+d}{c-d}$.
- Very useful in **algebra, trigonometry, and number theory proofs**.

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Do you want me to also show you **proof using numbers step by step** (like substituting values) so it becomes crystal clear?

Is this conversation helpful so far?